Task 2

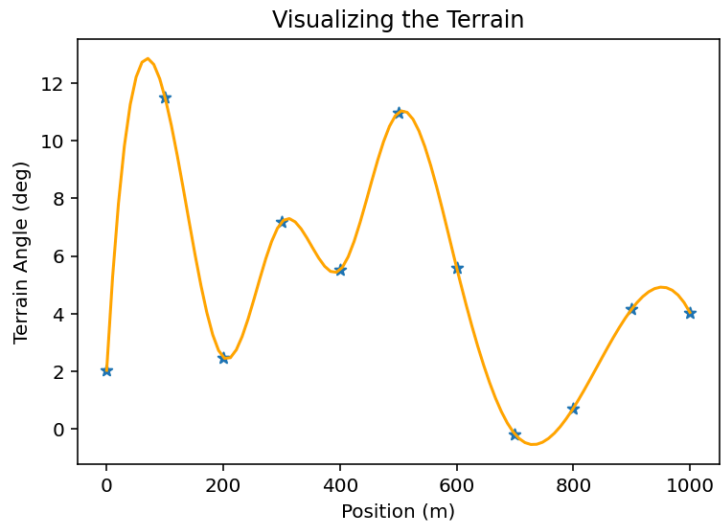

Figure 1: Mars terrain by angle of elevation by position

The figure above shows the two plots of the known data points shown as stars along with the interpolated terrain angles shown in the orange curve. It can be seen that the curve is smooth. This is because we used 100 position points over a total of 1000 meters that were then used to interpolate the terrain angle. A cubic polynomial was used during the interpolation process to ensure accuracy and prevent any discontinuities. In addition, the curve intersects all of the known data points, which provides further proof that the process was derived correctly.

Task 5

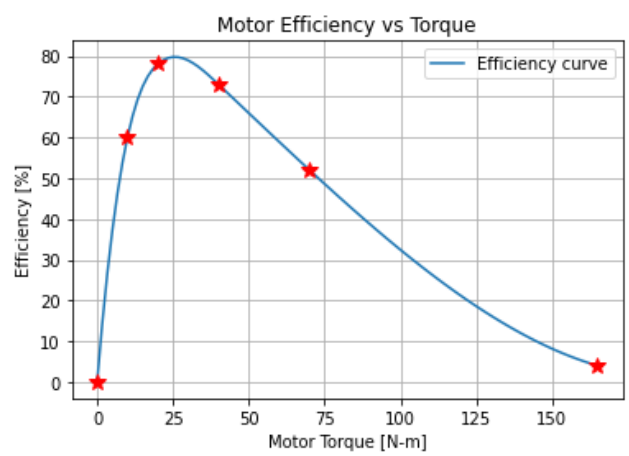

Figure 2: Motor Efficiency vs. Torque plot

The figure above shows a plot that displays the relationship between the motor efficiency and the motor torque, and the stars represent the given values provided in the dictionary. This graph illustrates that at low torque values (0 - 10 N-m), the efficiency of the motor is below 60%. This illustrates that during low torque, the motor is operating easily due to low torque, but not doing useful work. Due to this, at low

motor torque, energy is wasted and the motor is inefficient. Between (20 - 50 N-m), the motor is operating at its highest efficiency because that region is where maximum work output is highest relative to losses. However, at higher motor torque values, the work output is lower relative to the losses, which results in lower efficiency. The graph produced is similar to Figure 2 from the Project Phase 2 Manual, which also simulates a motor efficiency graph where the maximum efficiency is when the motor torque is at a moderate value. But at lower and higher motor torque values, similar to the plot above, the efficiency declines due to waste in work. The plot is "smooth" because the interpolation method and the theoretical physics of the motor produce a continuous behavior. The factors determining the efficiency of the motor, motor torque, changes continuously, and therefore there are no abrupt changes. The curve also goes through all the known data points because interpolation is used, which constructs a continuous function that adheres to the known values while estimating the values in between the known values.

Task 8

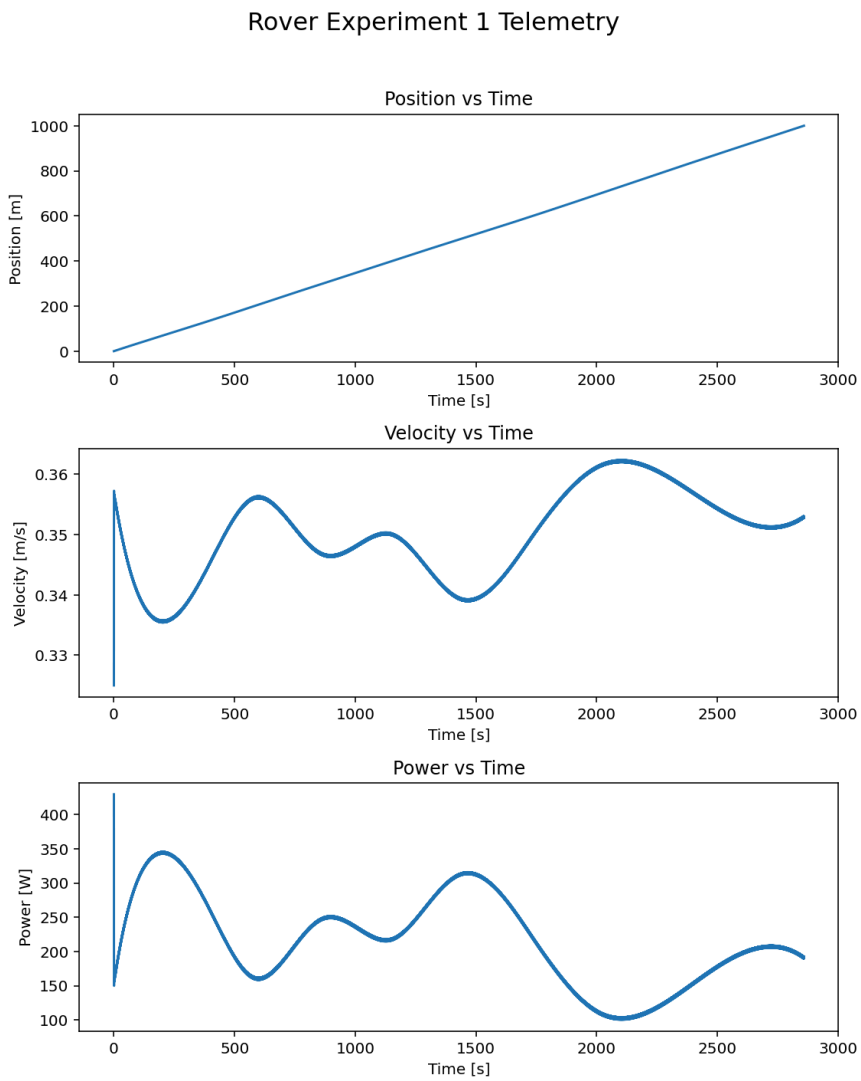

Figure 3: Position, Velocity and Power vs Time Graphs

Property	Units	Value
Completion Time	s	2859.48
Distance Traveled	m	1000
Max Velocity	m/s	0.36
Average Velocity	m/s	0.35
Battery Energy	KJ	1078
Energy Per Distance	J/m	1078.26

Figure 4: Table of Experimental Rover Properties

For the experiment the rover travels a kilometer at the speed of an average walk across the terrain which is pretty steady based on the angle of elevation of the terrain. Across the terrain the rover goes both up and downhill speeding up and down based on this change in elevation. This also correlates to the amount of power needed as at higher speeds going down hill less power is needed while uphill takes many more Watts to get back up to speed combined with the added forces to go uphill. With these relationships the velocity and power graphs oscillate on opposite cycles as the velocity increases less power is needed to keep the average speed relatively constant. It's like an intrinsic PID loop as they balance each other. As this speed is pretty constant only going from around .34 to .36m/s the position graph sees very little variation giving a roughly linear slope to the naked eye.

For this experiment the ending condition applied is the distance traveled hitting the 1000m limit set before a pause to reevaluate the rover. This is due to the rover not coming to a stop with any velocities under .01m/s and the distance being reached before 10,000 seconds elapse.

Task 9

The power was found by integrating the average velocity over the experiment and relating this is power with time and the rover's physical dimensions. This power added up to 1078 KJ over the 1000m trek. As the provided battery pack only provides 907 KJ of energy an alternative light-weight power source will need to be specced out under these conditions. I would also give a factor of safety for this power capacity as charging the solar panels while driving may not be feasible for the durability of the panels.